

The Foundations: Logic and Proofs

Chapter 2, Part I: Propositional Logic

Propositional Logic

LOGIC

Why logic in Discrete Structures?

Logic helps us:

- Decide whether a statement is **true or false**
- Check whether an **argument is valid**
- Design **conditions in programs (if–else, loops)**

Most Importantly, logic = **decision making**

PROPOSITIONS

- ▶ The basic building blocks of logic- propositions.
- ▶ A **proposition** is a declarative sentence (that is, a sentence that declares a fact) that is **either true or false, but not both**.
- ▶ The **truth value** of a proposition is true, denoted by T, if it is a true proposition, and the truth value of a proposition is false, denoted by F, if it is a false proposition.

Sentence	Proposition?	Reason
" $2 + 2 = 4$ "	Yes	True
" $2 + 2 = 5$ "	Yes	False
"Is it raining?"	No	Question
"Close the door."	No	Command
" $x + 1 = 3$ "	No	Depends on x

Key rule:

If you cannot assign **T** or **F**, it is **NOT** a proposition

Propositional Variables

- We use letters to represent propositions:
 - $p, q, r, s \dots$

Example:

- **p:** “It is raining”
- **q:** “I am at home”
- Each can be **T or F**

NEGATION OF A PROPOSITION

DEFINITION:

- Let p be a proposition. The *negation of p* , denoted by $\sim p$.
- The proposition $\sim p$ is read “not p .”
- The truth value of the negation of p , $\sim p$, is the opposite of the truth value of p .

REMARK: Other notations you might see are $\sim p$, $\neg p$, p' , Np , and $!p$.

EXAMPLE : Find the negation of the proposition

“Michael’s PC runs Linux” and express this in simple English.

SOLUTION:

“Michael’s PC does not run Linux.”

EXAMPLE 2 :

If p denotes “The earth is round.”, then

$\neg p$ denotes “It is not the case that the earth is round”
or more simply “The earth is not round.”

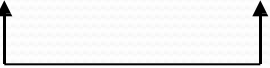
TABLE 1 The Truth Table for the Negation of a Proposition.

p	$\neg p$
T	F
F	T

DOUBLE NEGATION PROPERTY

$$\sim(\sim p) \equiv p$$

p	$\sim p$	$\sim(\sim p)$
T	F	T
F	T	F



EXAMPLE

“It is not true that I am not happy”

SOLUTION:

Let p = “I am happy”

then $\sim p$ = “I am not happy”

and $\sim(\sim p)$ = “It is not true that I am not happy”

Since $\sim(\sim p) \equiv p$

Hence the given statement is equivalent to:

“I am happy”

COMBINING PROPOSITIONS

- ▶ The logical operators that are used to form **new propositions from two or more existing propositions**.
- ▶ These logical operators are also called **Connectives**.

CONNECTIVE	MEANING	SYMBOL	CALLED
Negation	not	$\sim$	Tilde
Conjunction	and	$\wedge$	Hat
Disjunction	or	$\vee$	Vel
Exclusive Or	either	$\oplus$	Xor
Implication	If....then...	$\rightarrow$	Arrow
Biconditional	If and only if	$\leftrightarrow$	Bidirectional Arrow

1. CONJUNCTION

Let p and q be propositions. The *conjunction* of p and q , denoted by $p \wedge q$, is the proposition “ p and q .” The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

► **Example:** If p denotes “I am at home.” and q denotes “It is raining.” then $p \wedge q$ denotes “I am at home and it is raining.”

► **Example 2 :**

p : “I studied” and q : “I passed”

$p \wedge q$ = “I studied and I passed”

TABLE 2 The Truth Table for the Conjunction of Two Propositions.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

EXAMPLE

Find the conjunction of the propositions p and q where p is the proposition “Rebecca’s PC has more than 16 GB free hard disk space” and q is the proposition “The processor in Rebecca’s PC runs faster than 1 GHz.”

SOLUTION:

- The conjunction of these propositions, $p \wedge q$, is the proposition “Rebecca’s PC has more than 16 GB free hard disk space, **and the** processor in Rebecca’s PC runs faster than 1 GHz.”
- **OR**
- This conjunction can be expressed more simply as “Rebecca’s PC has more than 16 GB free hard disk space, **and its** processor runs faster than 1 GHz.”

For this conjunction to be true, both conditions given must be true. It is false when one or both of these conditions are false.

2. DISJUNCTION (INCLUSIVE OR)

Let p and q be propositions. The *disjunction* of p and q , denoted by $p \vee q$, is the proposition “ p or q .” **The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.**

- ▶ A disjunction is true when at least one of the two propositions is true. That is, $p \vee q$ is true when both p and q are true or when exactly one of p and q is true.

Example:

If p denotes “I am at home.”

and q denotes “It is raining.”

then $p \vee q$ denotes

“I am at home or it is raining.”

TABLE 3 The Truth Table for the Disjunction of Two Propositions.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

EXAMPLE

Translate the statement “Students who have taken calculus or introductory computer science can take this class” in a statement in propositional logic using the propositions p : “A student who has taken calculus can take this class” and q : “A student who has taken introductory computer science can take this class.”

SOLUTION:

We assume that this statement means that **students who have taken either one of the subject: calculus or introductory computer science can take the class.**

Hence, this statement can be expressed as $p \vee q$, the inclusive or, or disjunction, of p and q .

3. (EXCLUSIVE OR):

- ▶ Let p and q be propositions. The *exclusive or of p and q* , denoted by $p \oplus q$ (or p XOR q), is the proposition that is true when **exactly one of p and q is true and is false otherwise.**
- ▶ The truth table for the exclusive or of two propositions is displayed in Table 4.

TABLE 4 The Truth Table for the Exclusive Or of Two Propositions.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

EXAMPLE

- Express the statement “I will use all my savings to travel to Europe or to buy an electric car” in propositional logic using the statement p : “I will use all my savings to travel to Europe” and the statement q : “I will use all my savings to buy an electric car.”

SOLUTION:

To translate this statement, we first note that the or in this statement must be an exclusive because this student can either use all his or her savings to travel to Europe or use all these savings to buy an electric car, but cannot both go to Europe and buy an electric car. (This is clear because either option requires all his savings.) Hence, this statement can be expressed as $p \oplus q$.

The Connective Or in English

- In English “or” has two distinct meanings.
 - **“Inclusive Or”** - In the sentence “Students who have taken CS202 or Math120 may take this class,” we assume that students need to have taken one of the prerequisites, but may have taken both. This is the meaning of disjunction. **For $p \vee q$ to be true, either one or both of p and q must be true.**
 - **“Exclusive Or”** - When reading the sentence “Soup or salad comes with this dinner,” **we do not expect to be able to get both soup and salad. In $p \oplus q$, one of p and q must be true, but not both.**

TRANSLATING FROM ENGLISH TO SYMBOLS

Let p = “It is hot”, and q = “It is sunny”

SENTENCE

- ▶ 1. It is **not** hot.
- ▶ 2. It is hot **and** sunny.
- ▶ 3. It is hot **or** sunny.
- ▶ 4. It is **not** hot **but** sunny.
- ▶ 5. It is **neither** hot **nor** sunny.

SYMBOLIC FORM

$\sim p$

$p \wedge q$

$p \vee q$

$\sim p \wedge q$

$\sim p \wedge \sim q$

4. CONDITIONAL STATEMENTS-

IMPLICATION:

Let p and q be propositions. The *conditional statement* $p \rightarrow q$ is the proposition “**if p , then q .**” The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise.

In the conditional statement $p \rightarrow q$, p is called the ***hypothesis*** (or *antecedent* or *premise*) and q is called the ***conclusion*** (or *consequence*).

What does $p \rightarrow q$ mean?

It means:

“If p happens, then q must happen.”

Think of it as a **promise or rule**.

The ONE rule you must remember

$p \rightarrow q$ is FALSE

only when p happens but q does NOT happen.

In all other cases, it is TRUE.

TABLE 5 The Truth Table for the Conditional Statement

$p \rightarrow q$.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Implication

Example:

“If it rains, I will take an umbrella.”

- Rains + umbrella $\rightarrow$ TRUE
- Rains + no umbrella $\rightarrow$ FALSE
- No rain + umbrella $\rightarrow$ TRUE
- No rain + no umbrella $\rightarrow$ TRUE

Implication

Example:

“If a number is divisible by 4, then it is even.”

- $8 \rightarrow$ divisible by 4 & even $\rightarrow$ ✓
- $6 \rightarrow$ not divisible by 4 but even $\rightarrow$ ✓
- $9 \rightarrow$ not divisible by 4 & not even $\rightarrow$ ✓

Only false if:

- Divisible by 4 but NOT even (impossible)

A conditional statement $p \rightarrow q$ is false only when p is true and q is false.

Different Ways of Expressing $p \rightarrow q$

- **if p , then q**
- **if p , q**
- **q unless $\neg p$**
- **q if p**
- **q whenever p**
- **q follows from p**
- **p implies q**
- **p only if q**
- **q when p**
- **p is sufficient for q**
- **q is necessary for p**
- **a necessary condition for p is q**
- **a sufficient condition for q is p**

CONVERSE, CONTRAPOSITIVE, AND INVERSE

From a conditional statement $p \rightarrow q$

We can form **three related statements**:

Name	Form	Meaning
Converse	$q \rightarrow p$	Swap p and q
Inverse	$\neg p \rightarrow \neg q$	Negate both p and q
Contrapositive	$\neg q \rightarrow \neg p$	Swap and negate

Important fact:

The contrapositive is logically equivalent to the original statement.

EXAMPLE K

- ▶ Find the contrapositive, the converse, and the inverse of the conditional statement:

“If it is raining, then the home team wins.”

SOLUTION:

The contrapositive of this conditional statement is

- ▶ **“If the home team does not win, then it is not raining.”**

The converse is

- ▶ **“If the home team wins, then it is raining.”**

The inverse is

- ▶ **“If it is not raining, then the home team does not win.”**

Only the contrapositive is equivalent to the original statement.

BICONDITIONALS

DEFINITION:

Let p and q be propositions. The *biconditional statement* $p \leftrightarrow q$ is the proposition “ p if and only if q .” The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise.

Biconditional statements are also called *bi-implications*.

”

Note that $p \leftrightarrow q$ has exactly the same truth value as $(p \rightarrow q) \wedge (q \rightarrow p)$.

TABLE 6 The Truth Table for the Biconditional $p \leftrightarrow q$.		
p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example:

- If p denotes “I am at home.” and q denotes “It is raining.” then $p \leftrightarrow q$ denotes “I am at home if and only if it is raining.”
- Some alternative ways “ p if and only if q ” is expressed in English:
 - p is necessary and sufficient for q
 - if p then q , and conversely

EXAMPLE

Let p be the statement “You can take the flight,” and let q be the statement “You buy a ticket.” Then $p \leftrightarrow q$ is the statement

► “You can take the flight if and only if you buy a ticket.”

This statement is true if p and q are either both true or both false, that is, if you buy a ticket and can take the flight or if you do not buy a ticket and you cannot take the flight. It is false when p and q have opposite truth values, that is, when you do not buy a ticket, but you can take the flight (such as when you get a free trip) and when you buy a ticket but you cannot take the flight (such as when the airline bumps you).

Truth Tables For Compound Propositions

- Construction of a truth table:
- **Rows**
 - Rows show all possible true/false cases.
 - If there are n propositions, there are 2^n rows.
 - Example: 1 proposition $\rightarrow$ 2 rows and 2 propositions $\rightarrow$ 4 rows
- **Columns**
 - Columns show how the statement is evaluated step by step.
 - The last column is for the final compound proposition.

Example Truth Table (Compound Propositions)

- Construct a truth table for $p \vee q \rightarrow \neg r$

p	q	r	$\neg r$	$p \vee q$	$p \vee q \rightarrow \neg r$
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	T	F	T

Equivalent Propositions

- Two propositions are *equivalent* if they always have the same truth value.
- **Example:** Show using a truth table that the conditional is equivalent to the **contrapositive**.

Solution:

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Using a Truth Table to Show Non-Equivalence

Example: Show using truth tables that neither the **converse nor inverse** of an implication are not equivalent to the implication.

Solution:

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg p \rightarrow \neg q$	$q \rightarrow p$
T	T	F	F	T	T	T
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	T	T	T

Precedence of Logical Operators

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

$p \vee q \rightarrow \neg r$ is equivalent to $(p \vee q) \rightarrow \neg r$

If the intended meaning is $p \vee (q \rightarrow \neg r)$

then parentheses must be used, otherwise the meaning changes.

Without brackets, logic symbols follow fixed priority rules.

Use parentheses when you want a different meaning.

Applications of Propositional Logic

Section 2.2

Applications of Propositional Logic: Summary

- Translating English to Propositional Logic
- System Specifications
- Logic Circuits

Translating English Sentences

- Steps to convert an English sentence to a statement in propositional logic
 - Identify atomic propositions and represent using propositional variables.
 - Determine appropriate logical connectives
- “If I go to Harry’s or to the country, I will not go shopping.”
 - p : I go to Harry’s
 - q : I go to the country.
 - r : I will go shopping.

If p or q then not r .

$$(p \vee q) \rightarrow \neg r$$

Example

Problem: Translate the following sentence into propositional logic:

“You can access the Internet from campus only if you are a computer science major or you are not a freshman.”

● **Solution:** Let:

● a: You can access the Internet from campus

● c: You are a computer science major

● f: You are a freshman

● **Logical form:**

$$a \rightarrow (c \vee \neg f)$$

Important note:

“Only if” always translates to $\rightarrow$

System Specifications

- System and Software engineers take requirements in English and express them in a precise specification language based on logic.

Example: Express in propositional logic:

“The automated reply cannot be sent when the file system is full”

Solution: One possible solution: Let p denote “The automated reply can be sent” and q denote “The file system is full.”

$$q \rightarrow \neg p$$

Consistent System Specifications

Definition: A list of propositions is *consistent* if it is possible to assign truth values to the proposition variables so that **each proposition is true**.

OR

A set of logical statements is **consistent** if you can assign **True/False** to each variable so that **every single statement comes out True at the same time**.

Consistent System Specifications

Exercise: Are these specifications consistent?

- “The diagnostic message is stored in the buffer or it is retransmitted.”
- “The diagnostic message is not stored in the buffer.”
- “If the diagnostic message is stored in the buffer, then it is retransmitted.”

Solution: Let p denote “The diagnostic message is stored in the buffer.” Let q denote “The diagnostic message is retransmitted”

Three specifications:

$p \vee q$ (The message is stored OR retransmitted)

$\neg p$ (The message is not stored)

$p \rightarrow q$ (If stored, then retransmitted)

Consistent System Specifications

We test: Can we pick values for p and q that make all three true?

- From statement 2: $p = \text{False}$
- From statement 1: Since p is False, q must be True for $p \vee q$ to be True.
- Check statement 3: $p \rightarrow q$ is "False $\rightarrow$ True", which is True.
All three can be true with $p = \text{False}$, $q = \text{True}$, So it's **consistent**.

Inconsistent Example

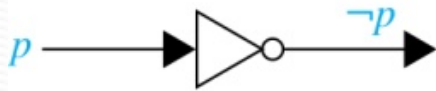
- Add a statement 4: $\neg q$ (The message is not retransmitted)
- Now:
- Statement 1: $p \vee q \rightarrow$ needs p or q **true**.
- Statement 2: $\neg p \rightarrow p$ **false**.
- Statement 4: $\neg q \rightarrow q$ **false**.
- But if p false and q false, statement 1 $p \vee q$ is false. 
There is **no possible assignment** to make all four true at once
 $\rightarrow$ **inconsistent**.

Consistent System Specifications

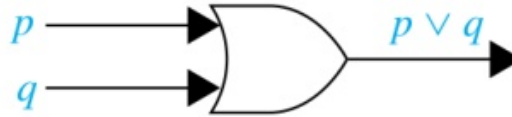
- **Real-World Use**
- Engineers check **consistency** to ensure system requirements don't contradict each other *before* building the system. If they're inconsistent, the system can't satisfy all requirements simultaneously.

Logic Circuits

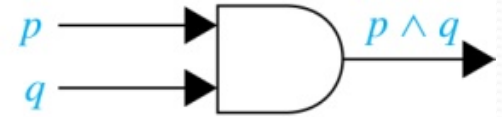
- **Electronic circuits**; each input/output signal can be viewed as a **0** or **1**.
 - **0** represents **False**
 - **1** represents **True**
- Complicated circuits are constructed from three basic circuits called **gates**.



Inverter



OR gate



AND gate

- The inverter (**NOT gate**) takes an input bit and produces the **negation of that bit**.
- The **OR gate** takes two input bits and produces the value **equivalent to the disjunction of the two bits**.
- The **AND gate** takes two input bits and produces the value **equivalent to the conjunction of the two bits**.

Logic Circuits

- More complicated **digital circuits** can be constructed by combining these **basic circuits** to produce the desired output given the input signals by building a circuit for each piece of the output expression and then combining them.
- **For example:**

